MP3: Numerical Accuracy Experiments

MP3 M1: Trajectory Divergence

Region: x ∈ [-0.7530, -0.7490], y ∈ [0.0990, 0.1030], τ = 0.01

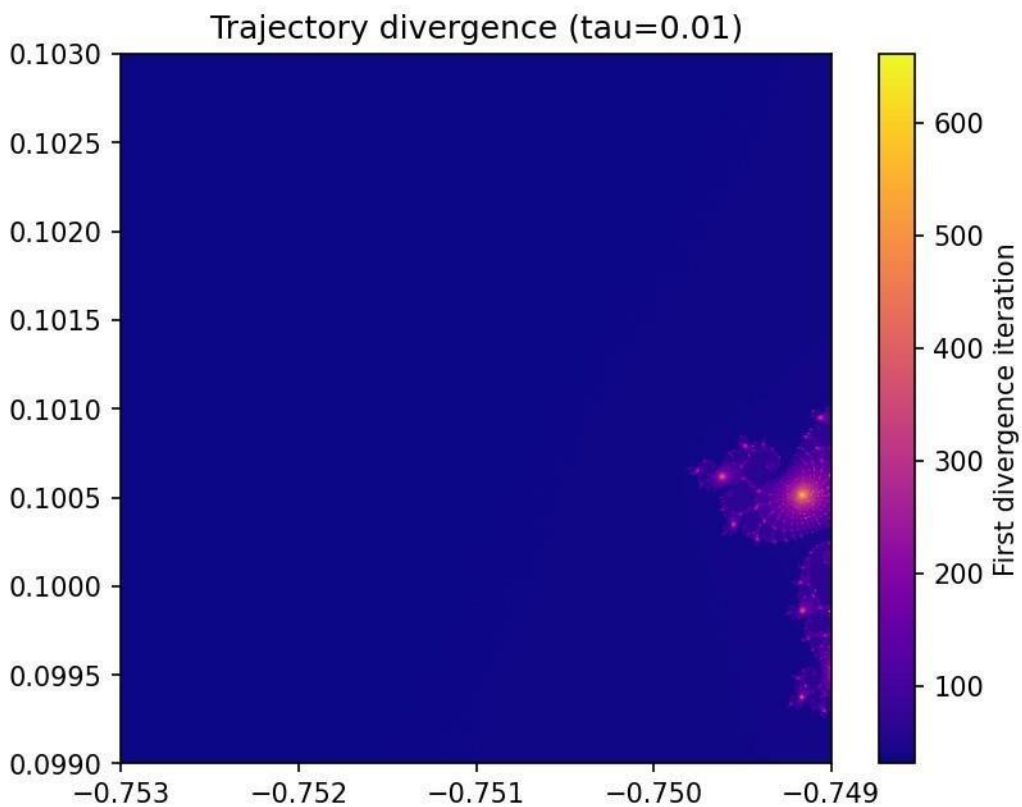

Figure 1: Trajectory divergence map (float32 vs float64) at Seahorse Valley

Observation:

Dark areas (early divergence) appear inside the Mandelbrot set and far outside.

Bright areas (late divergence) trace the fractal boundary (Seahorse Valley).

Float32 and float64 diverge exactly where the Mandelbrot set is most complex.

MP3 M2: Sensitivity Map

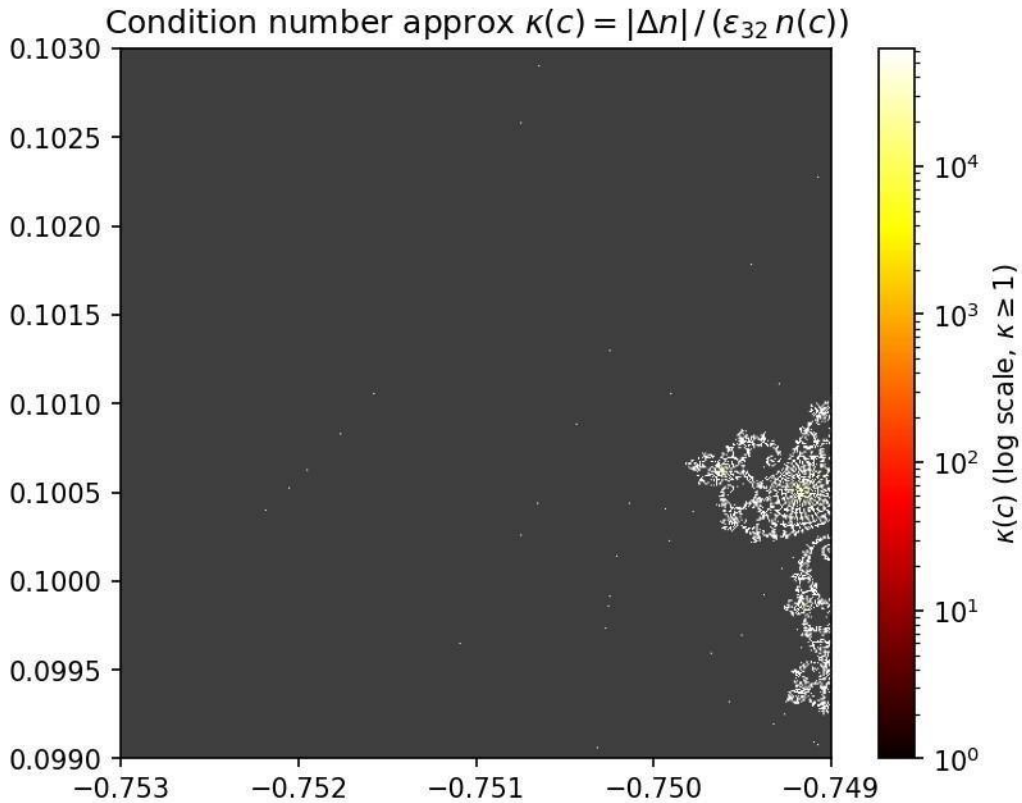

Figure 2: Sensitivity map (condition number approximation) at Seahorse Valley

Observation:

Dark areas (low κ) are well-conditioned (interior and exterior). Bright areas (high κ) are ill-conditioned (fractal boundary). The sensitivity map aligns with the trajectory divergence map: high sensitivity = late divergence = fractal boundary.

MP3 L09 M1: Test Suite

- Pass count: 5/5 tests passed
- Test coverage for test_mandelbrot.py: 100%
- Overall project coverage: 5% (only test file covered)

MP3 L10 M1: GPU Mandelbrot Results

- Device: Intel(R) HD Graphics 620
- Grid size: 1024×1024
- Max iterations: 100

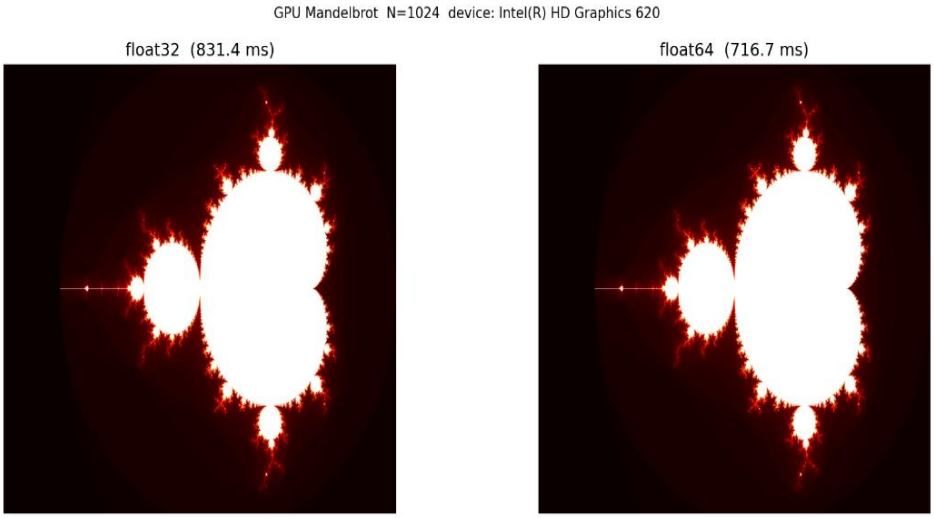

Figure 3: Side-by-side comparison of GPU Mandelbrot (float32 left, float64 right)

MP3 L10 M2: Float32 vs Float64 GPU Comparison

- float32 time: 0.831 seconds
- float64 time: 0.717 seconds
- Ratio (f64/f32): 0.86x
- Max pixel difference: 37 pixels

Comment: On this GPU, float64 was slightly faster than float32.